BIOVISION

SAMAGRA SHIKSHA KERALA — FIRST TERMINAL EVALUATION (2026-27)

STANDARD IX — PHYSICS (ANSWER KEY)

Answers | Time: 1 ½ Hours **Total Score: 40 | Code: 906-Ph**

SECTION A (1 Score each)

1. a. Both statement and reason are true and reason explains the statement.

2. b. i – r – y, ii – p – z, iii – q – x

3. c. Distance = 700 m, Displacement = 500 m

4. b. X-axis : 1 cm = 2 s, Y-axis : 1 cm = 10 m

SECTION B (2 Scores each)

5.

a) Non-uniform velocity

b) Uniform velocity

c) Non-uniform velocity

d) Uniform velocity

6.

- Pedestrians should walk along the right side of the road.
- Cross the road only through zebra crossings according to the traffic signals *(or wear light-coloured clothes while walking at night)*.

7.

a) Broken white lines on the road: Vehicles can cross or change lanes in a controlled manner.

b) Continuous yellow lines on the road: Vehicles should not cross or overtake across the line.

8.

a) 90° (48.6° is the critical angle of water).

b) When the angle of incidence is 60° (greater than the critical angle), Total Internal Reflection takes place. The light ray reflects back completely into the water at an angle of 60° without refracting into the air.

9.

a) Agree with the statement. When a body returns to its initial starting point after travel, the initial and final positions are the same; hence, the displacement is zero.

b) When an object travels in a straight line in the same direction, the distance travelled and the displacement will be equal.

10A.

a) Pour water into the vessel.

b) Scientific Principle: Refraction of Light. When water is poured, light rays coming from the coin travel from water to air, bending away from the normal and reaching the eye, making the coin appear slightly raised and visible.

— OR —

10B.

Light rays emerging from the fish in water bend away from the normal upon entering the air due to refraction. As a result, the fish appears slightly raised from its actual position. Therefore, bow fishermen aim slightly below the apparent position of the fish.

11A.

Due to Atmospheric Refraction. Even after the sun dips below the horizon, light rays from the sun pass through atmospheric layers of increasing optical density, undergo continuous refraction, and bend downwards into our eyes, making the sun visible for a short time above the horizon.

— OR —

11B.

a) Actual position: **B** (Apparent position: A)

b) Atmospheric Refraction. Due to variations in atmospheric density, light coming from the star continuously undergoes refraction and bends towards the normal, making the star appear higher than its actual position.

SECTION C (3 Scores each)

12.

- Velocity at t = 4 s, u = 5 m/s
- Velocity at t = 10 s, v = 35 m/s
- Time interval, t = 10 – 4 = 6 s
- Acceleration, a = (v – u) / t = (35 – 5) / 6 = 30 / 6 = **5 m/s^2**
- Displacement, s = ut + ½at^2 = (5 × 6) + (½ × 5 × 6^2) = 30 + 90 = **120 m**

13.

Refractive Index, n = c / v

- Water: n = (3 × 10^8) / (2.25 × 10^8) = **1.33**
- Glass: n = (3 × 10^8) / (2 × 10^8) = **1.5**
- Air: n = (3 × 10^8) / (3 × 10^8) = **1.0**

In Ascending Order: Air (1.0) < Water (1.33) < Glass (1.5)

14.

a) Figure **C**

b) Optical fibre, Vehicle reflectors, Periscope, Endoscopy (any two).

c) Justification: Total internal reflection occurs only when light travels from an optically denser medium to a rarer medium and the angle of incidence is greater than the critical angle of that pair of media.

15.

a) Speed from B for half an hour (0.5 h) covering 25 km = 25 / 0.5 = 50 km/h.
Since this is half the initial speed, the initial speed = 50 × 2 = 100 km/h.
Distance from A to B = Speed × Time = 100 km/h × 1 h = **100 km**.

b) Disagree with the opinion. Because here the speed is decreasing (it is retardation / non-uniform motion).

16A.

a) A coconut falling down from a coconut tree / A ball rolling down an inclined plane.

b) u = 0, s = 40 m, t = 7 s
$s = ut + \frac{1}{2}at^2 \Rightarrow 40 = 0 + (\frac{1}{2} \times a \times 7^2) \Rightarrow 40 = (49/2)\ a$
$a = 80 / 49 \approx$ **1.63 m/s^2**

— OR —

16B.

a) The rate of decrease of velocity is called negative acceleration or retardation.

b) u = 200 m/s, v = 0, t = 4 s
Acceleration, $a = (v - u) / t = (0 - 200) / 4 = -50$ m/s^2
Retardation = 50 m/s^2

17A.

a) Refraction of light.

b) Letters appear raised when a glass slab is placed over them / A coin in a water trough appears raised.

c) Light rays coming from the bottom of a water body bend away from the normal upon entering the air due to refraction, causing the bottom to appear raised.

— OR —

17B.

a) Refraction.

b) False. Vehicle reflectors work on the principle of 'Total Internal Reflection', whereas plane mirrors work on 'Regular Reflection'.

SECTION D (4 Scores)

18A.

a) Angle of incidence i = 90° – 40° = **50°**, Angle of refraction r = **40°**

b) Medium B has higher optical density than Medium A (since the refracted ray bends towards the normal, i > r).

c) At the bottom surface of the slab, the refracted ray bends away from the normal and emerges parallel to the incident ray (with lateral displacement).

— OR —

18B.

a) Conditions:

1. Light must travel from one medium to another with a difference in optical density.
2. The light ray must enter the interface obliquely (not perpendicularly).

b)

- When entering from air to water, the light ray bends towards the normal.
- When entering from water to glass, it bends further towards the normal (as glass has higher optical density than water).

Binu M